Quiz-1: Pick the question as per the last digit of your ID Number and answer **Q**uestion 0-9 as per the last digit of your ID

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1. **Question 0: Given the matrix**

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{pmatrix},$$

   find its reduced row echelon form.

2. **Question 1: For the matrix**

$$B = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 5 \\ 1 & 3 & 2 & 4 \\ 3 & 6 & 4 & 8 \end{pmatrix},$$

   find its reduced row echelon form .

3. **Question 2: Find the reduced row echelon form of the matrix**

$$C = \begin{pmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 5 \\ 3 & 6 & 4 & 8 \\ 4 & 8 & 5 & 9 \end{pmatrix}.$$

   Is this matrix invertible? Justify your answer.

4. **Question 3: Given the matrix**

$$D = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix},$$

find its reduced row echelon form .

5. **Question 4: Determine the reduced row echelon form of the matrix**

$$E = \begin{pmatrix} 2 & 1 & 3 & 1 \\ 4 & 3 & 6 & 5 \\ 6 & 5 & 9 & 7 \\ 8 & 7 & 12 & 10 \end{pmatrix}.$$

6. **Question 5: Find the reduced row echelon form of the matrix**

$$F = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 6 & 2 & 8 & 4 \\ 9 & 3 & 12 & 6 \\ 12 & 4 & 16 & 8 \end{pmatrix}.$$

7. **Question 6: Given the matrix**

$$G = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 1 & 6 & 7 \\ 0 & 1 & 8 & 9 \end{pmatrix},$$

find its reduced row echelon form.

8. **Question 7: For the matrix**

$$H = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix},$$

find the reduced row echelon form and verify if it is the identity matrix.

9. **Question 8: Given the matrix**

$$I = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

find its reduced row echelon form and discuss its implications for the solutions to the corresponding system of linear equations.

10. **Question 9: For the matrix**

$$J = \begin{pmatrix} 5 & 10 & 15 & 20 \\ 0 & 5 & 10 & 15 \\ 0 & 0 & 5 & 10 \\ 0 & 0 & 0 & 5 \end{pmatrix},$$

find the reduced row echelon form. Is the matrix invertible? Justify your answer.